

Aim:- To write SQL queries for implementing aggregate functions such as MAX(), MIN(), AVG() & COUNT().

Theory Aggregate functions in SQL are used to perform calculations on a set of values & return a single result.

They are mainly used with numeric data & often combined with the GROUP BY clause.

Types of Aggregate Function:-

1. MAX() → Returns the maximum value from a column.
2. MIN() → Returns the minimum value from a column.
3. AVG() → Returns the average value of a column.
4. COUNT() → Returns the number of rows.

Additional concept

GROUP BY → Groups rows with similar values.

HAVING → Used to filter grouped data.

Algorithm

1. Start SQL environment
2. Select the database
3. Use aggregate function on tables
4. Apply GROUP BY where required
5. Displays result.

Program:-

Step 1 Select Database
use college DB;

MAX Function -

Step 2 Find maximum salary of Teacher.
SELECT MAX (Salary) AS MAX-Salary
FROM Teacher.

MIN Function

Step 3 Find minimum salary of Teacher.
SELECT MIN (Salary) AS MIN-Salary
FROM Teacher;

AVG Function

Step 4 Find average salary of Teacher.
SELECT AVG (Salary) AS Avg-Salary
FROM Teacher;

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COUNT Function

Step 5 : Count Total Students

```
SELECT COUNT(*) AS Total_Students  
FROM Student;
```

GROUP BY Function

Step 6 : Count Student in each course

```
SELECT Course_ID, COUNT(*) AS Total_Students  
FROM Student  
GROUP BY Course_ID;
```

GROUP BY with Avg

Step 7 : Average Marks in Exam Result

```
SELECT Exam_ID, AVG(Total_Marks) AS Avg_marks  
FROM Exam_Result  
GROUP BY Exam_ID;
```

HAVING Clause

Step 8 : Courses with more than 1 student

```
SELECT Course_ID, COUNT(*) AS total_Students  
FROM Student  
GROUP BY Course_ID  
HAVING COUNT(*) > 1;
```

Output Table

Teacher Table		
Teacher-ID	None	Salary
1	Raj	40000
2	Amit	50000
3	Neha	60000

Min Salary = 40000

Max Salary = 60000

Arg salary = 50000

Total student = 3

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`SELECT * FROM Students;`
`SELECT * FROM Teacher;`

Result - Thus, SQL aggregate function MAX(), MIN(), AVG() & COUNT() were successfully implemented.

Viva Question:-

Q1 What are aggregate Function?
Ans Functions that perform calculations on multiple rows to return a single value.

Q2 Difference between COUNT(*) & COUNT(column)?
Ans COUNT(*) counts all rows
COUNT(column) ignores NULL values.

Q3 What is GROUP BY?
Ans Used to group rows having the same values.

Q4 What is HAVING?
Ans Used to filter grouped data.

Q5 Difference between WHERE & HAVING?
Ans WHERE → before grouping
HAVING → after grouping

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Aim - To write SQL queries for implementing Integrity Constraints in database tables.

Theory - Integrity constraints are rules applied on database tables to maintain the accuracy, consistency & reliability of data. These constraints prevent invalid data entry into the database.

Types of integrity Constraints -

1. PRIMARY KEY : Uniquely identifies each record in table
Cannot contain NULL values.

Ex

Student ID INT PRIMARY KEY

2. FOREIGN KEY :- Creates relationship between two tables.

Maintain referential integrity.

Ex

FOREIGN KEY (Course_id) REFERENCES Course (Course_id)

3. NOT NULL :- Ensures that a column cannot contain NULL values

Ex

Name VARCHAR(50) NOT NULL.

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4. UNIQUE Prevents duplicate values in a column.

Ex

Gmail VARCHAR(100) UNIQUE

5. CHECK restricts values based on a condition.

Ex Age INT CHECK (Age >= 18)

6. DEFAULT :- Assigns default value if no value is provide

Ex

Status VARCHAR(10) DEFAULT 'Active'

Algorithm

1. Start SQL Environment
2. Create database & select it.
3. Create tables using integrity Constraints
4. Insert record into tables
5. Display records
6. Verify constraints.

Program

Step 1 Create Database

```
CREATE Database CollegeDB;
```

Step 2 Select Database

```
USE CollegeDB;
```

Step 3 Create Course Table

```
CREATE TABLE COURSE (  
  COURSE_ID INT PRIMARY KEY,  
  Course_Name VARCHAR(100) NOT NULL,  
  Course_Fee INT CHECK (Course_Fee > 0),  
  Status VARCHAR(10) DEFAULT 'ACTIVE'  
);
```

Step 4 Create student Table with constraints.

```
CREATE TABLE Student (  
  Student ID INT PRIMARY KEY,  
  NAME VARCHAR(100) NOT NULL,  
  Gender VARCHAR(10),  
  Age INT CHECK (Age >= 10),  
  Email VARCHAR(100) UNIQUE,  
  Course ID INT,  
  FOREIGN KEY (Course ID)  
  REFERENCES Course (Course ID)  
);
```


Output

Course table

Course_id	Course_Name	Course_fee	Status
101	B.tech	50000	Active
102	BCA	40000	Active

Student table

Student-id	Name	Gender	Age	Email	Courseid
1	Rahul	Male	20	rahul@gmail.com	101
2	Priya	Female	19	priya@gmail.com	102

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Data Insertion -

Step 5 Insert data into Course table

```
INSERT INTO Course VALUES  
(101, 'B.tech', 50000, 'Active')  
(102, 'BCA', 40000, 'Active');
```

Step 6 insert data into Student table

```
INSERT INTO Student  
(Student-ID, Name, Gender, Age, Email,  
Course-ID)  
VALUES  
(1, 'Rahul', 'Male', 20, 'rahul@gmail.com',  
101),  
(2, 'Priya', 'Female', 19, 'priya@gmail.com',  
102);
```

Display records

Step 3

```
SELECT * FROM Course;  
SELECT * FROM Student;
```

Result: Thus, integrity constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, DEFAULT were successfully implemented in Database tables

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Viva Question

Q1. What are the integrity constraints?
Ans. Rules used to maintain Accuracy & consistency of data in a database

Q2. What is UNIQUE Constraint?
Ans. It prevents duplicate values in Column

Q3. What is CHECK Constraint?
Ans. It restricts value based on a specified ~~condition~~ column.

Q4. What is FOREIGN KEY?
Ans. A key used to establish relationship between two tables.

Q5. What is the PRIMARY KEY?
Ans. A key that uniquely identifies each record in a table.